

N1 Discrete probability distributions

Name: _____

Class: _____

Date: _____

Time: **431 minutes**

Marks: **354 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

Nicola, a darts player, is practising hitting the bullseye. She knows from previous experience that she has a probability of 0.3 of hitting the bullseye with each dart.

Nicola throws eight practice darts.

- (a) Using a binomial distribution, calculate the probability that she will hit the bullseye three or more times. (2)
- (b) Nicola throws eight practice darts on three different occasions. Calculate the probability that she will hit the bullseye three or more times on all three occasions. (2)
- (c) State two assumptions that are necessary for the distribution you have used in part (a) to be valid. (2)

(Total 6 marks)

Q2.

Abu visits his local hardware store to buy six light bulbs.

He knows that 15% of all bulbs at this store are faulty.

- (a) State a distribution which can be used to model the number of faulty bulbs he buys. (1)
- (b) Find the probability that all of the bulbs he buys are faulty. (1)
- (c) Find the probability that at least two of the bulbs he buys are faulty. (2)
- (d) Find the mean of the distribution stated in part (a). (1)
- (e) State two necessary assumptions in context so that the distribution stated in part (a) is valid. (2)

(Total 7 marks)

Q3.

The number of pots of yoghurt, X , consumed per week by adults in Milton is a discrete random variable with probability distribution given by

x	0	1	2	3	4	5	6	7 or more
$P(X = x)$	0.30	0.10	0.05	0.07	0.03	0.16	0.09	0.20

Find $P(3 \leq X < 6)$

Circle the correct answer.

0.26

0.31

0.35

0.40

(Total 1 mark)

Q4.

Ellie, a statistics student, read a newspaper article that stated that 20 per cent of students eat at least five portions of fruit and vegetables every day.

Ellie suggests that the number of people who eat at least five portions of fruit and vegetables every day, in a sample of size n , can be modelled by the binomial distribution $B(n, 0.20)$.

- (a) There are 10 students in Ellie's statistics class.

Using the distributional model suggested by Ellie, find the probability that, of the students in her class:

- (i) two or fewer eat at least five portions of fruit and vegetables every day; (1)

- (ii) at least one but fewer than four eat at least five portions of fruit and vegetables every day. (2)

- (b) Ellie's teacher, Declan, believes that more than 20 per cent of students eat at least five portions of fruit and vegetables every day. Declan asks the 25 students in his other statistics classes and 8 of them say that they eat at least five portions of fruit and vegetables every day.

- (i) Name the sampling method used by Declan. (1)

- (ii) Describe **one** weakness of this sampling method. (1)

- (iii) Assuming that these 25 students may be considered to be a random sample, carry out a hypothesis test at the 5% significance level to investigate whether Declan's belief is supported by this evidence. (6)

(Total 11 marks)

Q5.

A survey has found that, of the 2400 households in Growmore, 1680 eat home-grown fruit and vegetables.

- (a) Using the binomial distribution, find the probability that, out of a random sample of 25 households in Growmore, exactly 22 eat home-grown fruit and vegetables. (2)

- (b) Give a reason why you would **not** expect your calculation in part (a) to be valid for the 25 households in Gifford Terrace, a residential road in Growmore.

(1)

(Total 3 marks)

Q6.

Terence owns a local shop. His shop has three checkouts, at least one of which is always staffed.

A regular customer observed that the probability distribution for N , the number of checkouts that are staffed at any given time during the spring, is

$$P(N = n) = \begin{cases} \frac{3}{4} \left(\frac{1}{4}\right)^{n-1} & \text{for } n = 1, 2 \\ k & \text{for } n = 3 \end{cases}$$

- (a) Find the value of k .

(1)

- (b) Find the probability that a customer, visiting Terence's shop during the spring, will find at least 2 checkouts staffed.

(2)

(Total 3 marks)

Q7.

Stopoff owns a chain of hotels. Guests are presented with the bills for their stays when they check out.

The number of bills that contain errors may be modelled by a binomial distribution with parameters n and p , where $p = 0.30$.

Determine the probability that, in a random sample of 40 bills:

- (a) at most 10 bills contain errors;
(b) at least 15 bills contain errors;
(c) at least 6 bills but at most 18 bills contain errors.

(Total 7 marks)

Q8.

The volume, X millilitres, of energy drink in a bottle can be modelled by a normal random variable with mean 507.5 and standard deviation 4.0.

- (a) Find:

- (i) $P(X < 515)$;

(ii) $P(500 < X < 515)$;

(iii) $P(X \neq 507.5)$.

(5)

(b) Determine the value of x such that $P(X < x) = 0.96$.

(3)

(c) The energy drink is sold in packs of 6 bottles. The bottles in each pack may be regarded as a random sample.

Calculate the probability that the volume of energy drink in at least 5 of the 6 bottles in a pack is between 500 ml and 515 ml.

(3)

(Total 11 marks)

Q9.

An auction house offers items of jewellery for sale at its public auctions. Each item has a reserve price which is less than the lower price estimate which, in turn, is less than the upper price estimate. The outcome for any item is independent of the outcomes for all other items.

The auction house has found, from past records, the following probabilities for the outcomes of items of jewellery offered for sale.

Outcome	Probability
Item does not achieve its reserve price	0.15
Item achieves at least its reserve price	0.85
Item achieves at least its lower price estimate	0.50
Item achieves at least its upper price estimate	0.175

For example, the probability that an item achieves at least its lower price estimate but not its upper price estimate is 0.325.

A particular auction includes exactly 40 items of jewellery that may be assumed to be a random sample of such items.

(a) Use binomial distributions to find the probability that:

(i) at most 10 items do not achieve their reserve prices;

(1)

(ii) 25 or more items achieve at least their lower price estimates;

(2)

- (iii) exactly 2 items achieve at least their upper price estimates; (2)
- (iv) more than 10 items but fewer than 15 items achieve at least their reserve prices but not their lower price estimates. (4)
- (b) How many of the 40 items of jewellery would you expect to achieve at least their reserve prices but not their upper price estimates? (2)
- (Total 11 marks)**

Q10.

The records at a passport office show that, on average, 15 per cent of photographs that accompany applications for passport renewals are unusable.

Assume that exactly one photograph accompanies each application.

- (a) Determine the probability that in a random sample of 40 applications:
- (i) exactly 6 photographs are unusable;
 - (ii) at most 5 photographs are unusable;
 - (iii) more than 5 but fewer than 10 photographs are unusable. (7)
- (b) Calculate the mean and the standard deviation for the number of photographs that are unusable in a random sample of **32** applications. (3)
- (c) Mr Stickler processes 32 applications each day. His records for the previous 10 days show that the numbers of photographs that he deemed unusable were

8 6 10 7 9 7 8 9 6 7

By calculating the mean and the standard deviation of these values, comment, with reasons, on the suitability of the $B(32, 0.15)$ model for the number of photographs deemed unusable each day by Mr Stickler.

(4)
(Total 14 marks)

Q11.

A bank issues three versions of its credit card: classic, gold and platinum. A customer's application for a credit card may be refused because of the customer's poor credit rating.

- (a) The proportion of customers who are refused the classic version of the bank's credit card is 0.275.

Calculate the probability that, from a random sample of 10 customers applying for the classic version, exactly 2 applications are refused.

(3)

- (b) The proportion of customers who are refused the gold version of the bank's credit card is 0.65.

Determine the probability that, from a random sample of 40 customers applying for the gold version, the number of applications that are **accepted** is:

- (i) no more than 15;

(2)

- (ii) at least 10 but at most 20.

(3)

- (c) The proportion of customers who are refused the platinum version of the bank's credit card is 0.85.

Determine the probability that, from a random sample of 50 customers applying for the platinum version, more than 40 applications are refused.

(3)

(Total 11 marks)

Q12.

A general store sells lawn fertiliser in 2.5 kg bags, 5 kg bags and 10 kg bags.

- (a) The actual weight, W kilograms, of fertiliser in a 2.5 kg bag may be modelled by a normal random variable with mean 2.75 and standard deviation 0.15.

Determine the probability that the weight of fertiliser in a 2.5 kg bag is:

- (i) less than 2.8 kg;

- (ii) more than 2.5 kg.

(5)

- (b) The actual weight, X kilograms, of fertiliser in a 5 kg bag may be modelled by a normal random variable with mean 5.25 and standard deviation 0.20.

- (i) Show that $P(5.1 < X < 5.3) = 0.372$, correct to three decimal places.

(2)

- (ii) A random sample of **four** 5 kg bags is selected. Calculate the probability that none of the four bags contains between 5.1 kg and 5.3 kg of fertiliser.

(2)

(Total 9 marks)

Q13.

A bin contains a very large number of paper clips of different colours. The proportion of each colour is shown in the table.

Colour	White	Yellow	Green	Blue	Red	Purple
Proportion	0.15	0.15	0.20	0.15	0.22	0.13

- (a) Packets are filled from the bin. Each filled packet contains exactly 30 paper clips which may be considered to be a random sample.

Use binomial distributions to determine the probability that a filled packet contains:

- (i) exactly 2 purple paper clips; (3)
- (ii) a **total** of more than 10 red or purple paper clips; (3)
- (iii) at least 5 but at most 10 green paper clips. (3)

- (b) Jumbo packets are also filled from the bin. Each filled jumbo packet contains exactly 100 paper clips.

- (i) Assuming that the number of paper clips in a jumbo packet may be considered to be a random sample, calculate the mean and the variance of the number of **red** paper clips in a filled jumbo packet. (2)
- (ii) It is claimed that the proportion of red paper clips in the bin is greater than 0.22 and that jumbo packets do not contain random samples of paper clips.

An analysis of the number of red paper clips in each of a random sample of 50 filled jumbo packets resulted in a mean of 22.1 and a standard deviation of 4.17.

Comment, with numerical justification, on **each** of the two claims.

(3)

(Total 14 marks)

Q14.

Clay pigeon shooting is the sport of shooting at special flying clay targets with a shotgun.

- (a) Rhys, a novice, uses a single-barrelled shotgun. The probability that he hits a target is 0.45, and may be assumed to be independent from target to target.

Determine the probability that, in a series of shots at 15 targets, he hits:

- (i) at most 5 targets; (1)
- (ii) more than 10 targets; (2)
- (iii) exactly 6 targets;

(2)

(iv) at least 5 but at most 10 targets.

(3)

- (b) Sasha, an expert, uses a double-barrelled shotgun. She shoots at each target with the gun's first barrel and, only if she misses, does she then shoot at the target with the gun's second barrel.

The probability that she hits a target with a shot using her gun's first barrel is 0.85. The conditional probability that she hits a target with a shot using her gun's second barrel, given that she has missed the target with a shot using her gun's first barrel, is 0.80. Assume that Sasha's shooting is independent from target to target.

- (i) Show that the probability that Sasha hits a target is 0.97.

(2)

- (ii) Determine the probability that, in a series of shots at 50 targets, Sasha hits at least 48 targets.

(3)

- (iii) In a series of shots at 80 targets, calculate the mean number of times that Sasha shoots at targets with her gun's second barrel.

(2)

(Total 15 marks)

Q15.

An amateur tennis club purchases tennis balls that have been used previously in professional tournaments.

The probability that each such ball fails a standard bounce test is 0.15.

The club purchases boxes each containing 10 of these tennis balls. Assume that the 10 balls in any box represent a random sample.

- (a) Determine the probability that the number of balls in a box which fail the bounce test is:

- (i) at most 2;

(1)

- (ii) at least 2;

(2)

- (iii) more than 1 but fewer than 5.

(3)

- (b) Determine the probability that, in **5 boxes**, the total number of balls which fail the bounce test is:

- (i) more than 5;

(2)

- (ii) at least 5 but at most 10.

(3)

- (c) Calculate the mean and the variance for the total number of balls in **50 boxes** which fail the bounce test.

(2)

(Total 13 marks)

Q16.

The diameter, D millimetres, of an American pool ball may be modelled by a normal random variable with mean 57.15 and standard deviation 0.04.

- (a) Determine:

- (i) $P(D < 57.2)$;

(3)

- (ii) $P(57.1 < D < 57.2)$.

(2)

- (b) A box contains 16 of these pool balls. Given that the balls may be regarded as a random sample, determine the probability that all 16 balls have diameters less than 57.2 mm;

(2)

(Total 7 marks)

Q17.

An amateur tennis club purchases tennis balls that have been used previously in professional tournaments.

The probability that each such ball fails a standard bounce test is 0.15.

The club purchases boxes each containing 10 of these tennis balls. Assume that the 10 balls in any box represent a random sample.

- (a) Determine the probability that the number of balls in a box which fail the bounce test is:

- (i) at most 2;

(1)

- (ii) at least 2;

(2)

- (iii) more than 1 but fewer than 5.

(3)

- (b) Determine the probability that, in **5 boxes**, the total number of balls which fail the bounce test is:

- (i) more than 5;

(2)

- (ii) at least 5 but at most 10.

(3)

(Total 11 marks)

Q18.

The probability that Barry's cat, Sylvester, chooses to stay outside all night is 0.35, and the cat's choice is independent from night to night.

- (a) Determine the probability that, during a period of 2 weeks (14 nights), Sylvester chooses to stay outside:

- (i) on at most 7 nights;

(2)

- (ii) on at least 11 nights;

(2)

- (iii) on more than 5 nights but fewer than 10 nights.

(3)

- (b) Calculate the probability that, during a period of **3 weeks**, Sylvester chooses to stay outside on exactly 4 nights.

(3)

(Total 10 marks)

Q19.

Draught excluder for doors and windows is sold in rolls of nominal length 10 metres.

The actual length, X metres, of draught excluder on a roll may be modelled by a normal distribution with mean 10.2 and standard deviation 0.15.

- (a) Determine:

- (i) $P(X < 10.5)$;

(3)

- (ii) $P(10.0 < X < 10.5)$.

(3)

- (b) A customer randomly selects six 10-metre rolls of the draught excluder.

Calculate the probability that all six rolls selected contain more than 10 metres of draught excluder.

(3)

(Total 9 marks)

Q20.

During the winter, the probability that Barry's cat, Sylvester, chooses to stay outside all

night is 0.35, and the cat's choice is independent from night to night.

- (a) Determine the probability that, during a period of 2 weeks (14 nights) in winter, Sylvester chooses to stay outside:

(i) on at most 7 nights;

(2)

(ii) on at least 11 nights;

(2)

(iii) on more than 5 nights but fewer than 10 nights.

(3)

- (b) Calculate the probability that, during a period of **3 weeks** in winter, Sylvester chooses to stay outside on exactly 4 nights.

(3)

- (c) Barry claims that, during the summer, the number of nights per week, S , on which Sylvester chooses to stay outside can be modelled by a binomial distribution with

$$n = 7 \text{ and } p = \frac{5}{7}.$$

(i) Assuming that Barry's claim is correct, find the mean and the variance of S .

(2)

(ii) For a period of 13 weeks during the summer, the number of nights per week on which Sylvester chose to stay outside had a mean of 5 and a variance of 1.5.

Comment on Barry's claim.

(2)

(Total 14 marks)

Q21.

The proportion of members of a health club who play tennis is 0.35.

A sample of 40 members of the health club is selected at random.

Determine the probability that the number of these members who play tennis is:

(a) at most 12;

(2)

(b) at least 10 but at most 15;

(3)

(c) exactly 15.

(2)

(Total 7 marks)

Q22.

Each day, Margot completes the crossword in her local morning newspaper. Her completion times, X minutes, can be modelled by a normal random variable with a mean of 65 and a standard deviation of 20.

(a) Determine:

(i) $P(X < 90)$;

(ii) $P(X > 60)$.

(5)

(b) Given that Margot's completion times are independent from day to day, determine the probability that, during a particular period of 6 days:

(i) she completes one of the six crosswords in exactly 60 minutes;

(1)

(ii) she completes each crossword in less than 60 minutes;

(3)

(Total 9 marks)

Q23.

In a certain country, 15 per cent of the male population is left-handed.

(a) Determine the probability that, in a random sample of 50 males from this country:

(i) at most 10 are left-handed;

(2)

(ii) at least 5 are left-handed;

(2)

(iii) more than 6 but fewer than 12 are left-handed.

(3)

(b) In the same country, 11 per cent of the female population is left-handed.

Calculate the probability that, in a random sample of 35 females from this country, exactly 4 are left-handed.

(3)

(c) A sample of 2000 people is selected at random from the population of the country. The proportion of males in the sample is 52 per cent.

How many people in the sample would you expect to be left-handed?

(4)

(Total 14 marks)

Q24.

As part of her breakfast duties, Fatima, a newly-appointed trainee chef, fries eggs.

It may be assumed that, for a newly-appointed trainee chef, the proportion of egg yolks broken during frying is 0.15, and that the breaking of yolks is independent from egg to egg.

- (a) Determine the probability that, on a day when Fatima fries 50 eggs for breakfast, the number of yolks that she breaks is:

(i) exactly 6;

(3)

(ii) more than 6 but at most 12.

(3)

- (b) Calculate the mean and the variance for the number of yolks broken on a day when Fatima fries **80** eggs.

(2)

(Total 8 marks)

Q25.

The proportion of passengers who use senior citizen bus passes to travel into a particular town on 'Park & Ride' buses between 9.30 am and 11.30 am on weekdays is 0.45.

It is proposed that, when there are n passengers on a bus, a suitable model for the number of passengers using senior citizen bus passes is the distribution $B(n, 0.45)$.

- (a) Assuming that this model applies to the 10.30 am weekday 'Park & Ride' bus into the town:

(i) calculate the probability that, when there are **16** passengers, exactly 3 of them are using senior citizen bus passes;

(3)

(ii) determine the probability that, when there are **25** passengers, fewer than 10 of them are using senior citizen bus passes;

(2)

(iii) determine the probability that, when there are **40** passengers, at least 15 but at most 20 of them are using senior citizen bus passes;

(3)

(iv) calculate the mean and the variance for the number of passengers using senior citizen bus passes when there are **50** passengers.

(2)

- (b) (i) Give a reason why the proposed model may not be suitable.

(1)

(ii) Give a **different** reason why the proposed model would not be suitable for the number of passengers using senior citizen bus passes to travel into the town on the **7.15 am** weekday 'Park & Ride' bus.

(1)

(Total 12 marks)

Q26.

Mr Alott and Miss Fewer work in a postal sorting office.

- (a) The number of letters per batch, R , sorted incorrectly by Mr Alott when sorting batches of 50 letters may be modelled by the distribution $B(50, 0.15)$.

Determine:

- (i) $P(R < 10)$;
(ii) $P(5 \leq R \leq 10)$.

(4)

- (b) It is assumed that the probability that Miss Fewer sorts a letter incorrectly is 0.06, and that her sorting of a letter incorrectly is independent from letter to letter.

- (i) Calculate the probability that, when sorting a batch of **22** letters, Miss Fewer sorts exactly 2 letters incorrectly.

(3)

- (ii) Calculate the probability that, when sorting a batch of **35** letters, Miss Fewer sorts at least 1 letter incorrectly.

(2)

- (iii) Calculate the mean and the variance for the number of letters sorted **correctly** by Miss Fewer when she sorts a batch of **120** letters.

(2)

- (iv) Miss Fewer sorts a random sample of 20 batches, each containing 120 letters. The number of letters sorted **correctly** per batch has a mean of 112.8 and a variance of 56.86.

Comment on the assumptions that the probability that Miss Fewer sorts a letter incorrectly is 0.06, and that her sorting of a letter incorrectly is independent from letter to letter.

(3)

(Total 14 marks)

Q27.

A travel agency in Tunisia offers customers a 3-day tour into the Sahara desert by either coach or minibus.

- (a) The agency accepts bookings from 50 customers for seats on the coach. The probability that a customer, who has booked a seat on the coach, will **not** turn up to claim the seat is 0.08, and may be assumed to be independent of the behaviour of other customers.

Determine the probability that, of the customers who have booked a seat on the coach:

- (i) two or more will **not** turn up;

- (ii) three or more will **not** turn up.

(4)

- (b) The agency accepts bookings from 15 customers for seats on the minibus. The probability that a customer, who has booked a seat on the minibus, will **not** turn up to claim the seat is 0.025, and may be assumed to be independent of the behaviour of other customers.

Calculate the probability that, of the customers who have booked a seat on the minibus:

- (i) all will turn up;
- (ii) one or more will **not** turn up.

(4)

- (c) The coach has 48 seats and the minibus has 14 seats. If 14 or fewer customers who have booked seats on the minibus turn up, they will be allocated a seat on the minibus. If all 15 customers who have booked seats on the minibus turn up, one will be allocated a seat on the coach. This will leave only 47 seats available for the 50 customers who have booked seats on the coach.

Use your results from parts (a) and (b) to calculate the probability that there will be seats available on the coach for all those who turn up having booked such seats.

(4)

(Total 12 marks)

Q28.

For the adult population of the UK, 35 per cent of men and 29 per cent of women do not wear glasses or contact lenses.

- (a) Determine the probability that, in a random sample of 40 men, at most 15 do not wear glasses or contact lenses.

(3)

- (b) Calculate the probability that, in a random sample of 10 women, exactly 3 do not wear glasses or contact lenses.

(3)

- (c) (i) Calculate the mean and the variance for the number who **do** wear glasses or contact lenses in a random sample of 20 women.

(3)

- (ii) The numbers wearing glasses or contact lenses in 10 groups, each of 20 women, had a mean of 16.5 and a variance of 2.50.

Comment on the claim that these 10 groups were **not** random samples.

(3)

(Total 12 marks)

Q29.

For the adult population of the UK, 35 per cent of men and 29 per cent of women do not wear glasses or contact lenses.

- (a) Determine the probability that, in a random sample of 40 men:
- (i) at most 15 do not wear glasses or contact lenses; (3)
 - (ii) more than 10 but fewer than 20 do not wear glasses or contact lenses. (3)
- (b) Calculate the probability that, in a random sample of 10 women, exactly 3 do not wear glasses or contact lenses. (3)
- (c) (i) Calculate the mean and the variance for the number who **do** wear glasses or contact lenses in a random sample of 20 women. (3)
- (ii) The numbers wearing glasses or contact lenses in 10 groups, each of 20 women, had a mean of 16.5 and a variance of 2.50.
- Comment on the claim that these 10 groups were **not** random samples. (3)
- (Total 15 marks)

Q30.

A manufacturer produces three models of washing machine: basic, standard and deluxe. An analysis of warranty records shows that 25% of faults are on basic machines, 60% are on standard machines and 15% are on deluxe machines.

For basic machines, 30% of faults reported during the warranty period are electrical, 50% are mechanical and 20% are water-related.

For standard machines, 40% of faults reported during the warranty period are electrical, 45% are mechanical and 15% are water-related.

For deluxe machines, 55% of faults reported during the warranty period are electrical, 35% are mechanical and 10% are water-related.

- (a) Draw a tree diagram to represent the above information. (3)
- (b) Hence, or otherwise, determine the probability that a fault reported during the warranty period:
- (i) is electrical; (2)
 - (ii) is on a deluxe machine, given that it is electrical. (2)
- (c) A random sample of 10 electrical faults reported during the warranty period is

selected. Calculate the probability that exactly 4 of them are on deluxe machines.

(3)

(Total 10 marks)

Q31.

The post office in a market town is located within a small supermarket.

The probability that an individual customer entering the supermarket requires a service from:

- the post office only is 0.48;
- the supermarket only is 0.30;
- both the post office and the supermarket is 0.22.

It may be assumed that the service required is independent from customer to customer.

- (a) For a random sample of 12 individual customers, calculate the probability that exactly 5 of them require a service from the post office only.
- (b) For a random sample of 40 individual customers, determine the probability that more than 10 but fewer than 15 of them require a service from the supermarket only.
- (c) For a random sample of 100 individual customers, calculate the mean and the standard deviation for the number of them requiring a service from both the post office and the supermarket.

(3)

(3)

(3)

(Total 9 marks)

Q32.

A hotel has 50 single rooms, 16 of which are on the ground floor. The hotel offers guests a choice of a full English breakfast, a continental breakfast or no breakfast. The probabilities of these choices being made are 0.45, 0.25 and 0.30 respectively. It may be assumed that the choice of breakfast is independent from guest to guest.

- (a) On a particular morning there are 16 guests, each occupying a single room on the ground floor. Calculate the probability that exactly 5 of these guests require a full English breakfast.
- (b) On a particular morning when there are 50 guests, each occupying a single room, determine the probability that:
- (i) at most 12 of these guests require a continental breakfast;
- (ii) more than 10 but fewer than 20 of these guests require no breakfast.
- (c) When there are 40 guests, each occupying a single room, calculate the mean and

(3)

(2)

(3)

the standard deviation for the number of these guests requiring breakfast.

(4)

(Total 12 marks)

Q33.

Payton has a pay-as-you-go internet account. To save money, he reads his e-mail messages just after 6 pm each day. The probability that he has no e-mail messages to read at this time is 0.45, and the number of e-mail messages he receives is independent from day to day.

- (a) Calculate the probability that Payton has no e-mail messages to read on exactly 3 days during a 7-day period.

(3)

- (b) Determine the probability that, during June (30 days), Payton has no e-mail messages to read:

- (i) on fewer than 15 days;

(2)

- (ii) on more than 10 days;

(2)

- (iii) on at least 12 but at most 18 days.

(3)

(Total 10 marks)

Q34.

Each weekday, Monday to Friday, Trina catches a train from her local station. She claims that the probability that the train arrives on time at the station is 0.4, and that the train's arrival time is independent from day to day.

- (a) Assuming her claims to be true, determine the probability that the train arrives on time at the station:

- (i) on at most 3 days during a 2-week period (10 days);

(2)

- (ii) on more than 10 days but fewer than 20 days during an 8-week period.

(3)

- (b) (i) Assuming Trina's claims to be true, determine the mean and standard deviation for the number of times during a week (5 days) that the train arrives on time at the station.

(3)

- (ii) Each week, for a period of 13 weeks, Trina's travelling colleague, Suzie, records the number of times that the train arrives on time at the station. Suzie's results are

2 2 4 1 2 3 3 2 2 0 3 2 0

Calculate the mean and standard deviation of these values.

(3)

(iii) Hence comment on the likely validity of Trina's claims.

(2)

(Total 13 marks)

Q35.

The table shows, for a particular population, the proportion of people in each of the four main blood groups.

Blood group	O	A	B	AB
Proportion	0.40	0.28	0.20	0.12

(a) A random sample of 20 people is selected from this population.

Determine the probability that the sample contains:

(i) at most 10 people with blood group O;

(2)

(ii) exactly 3 people with blood group A;

(3)

(iii) more than 4 but fewer than 8 people with blood group B.

(3)

(b) A random sample of 500 people is selected from this population.

Find values for the mean and variance of the number of people in the sample with blood group AB.

(2)

(Total 10 marks)

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